MILITARY SPECIFICATION
DELAY ELEMENT, CARTRIDGE: XM44, ASSEMBLY

1. SCOPE

1.1 The specification covers Delay Element, Cartridge: XM44 Assembly, for use in cartridges for propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date on invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-B-550	- Barium Chromate
MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-P-46994	- Pellets, Boron, Potassium Nitrate
MIL-B-51092	- Boron, Amorphous Powder
MIL-C-60539	- Coating, Anodic, Conventional For Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions

FSC 1377

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2.1 (Cont'd)

DRAWINGS

Munitions Command

D10522708	- Delay Element, Cartridge: XM44 Assembly
C8864162	- Ammunition Container, Metal M21 for XM44 Delay Element
C8826729	- Box, Packing, Ammunition for Delay Element M5 or M8
IEL-10522708	- Inspection Equipment List for Delay Element, Cartridge: XM44, Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA-B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016)

3. REQUIREMENTS

3.1 General.-Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.-Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.-The delay element assembly shall withstand vibrations as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer or delay charge or derangement of components.

3.4 Functional.

3.4.1 The delay element shall not fail to fire.

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3.4.2 Ballistics.-The delay charge shall consist of pellets, boron, potassium nitrate as in specification MIL-P-46994, Barium Chromate as specified in MIL-B-550 and Boron as specified in MIL-B-51092 in the proportion and quantity fixed by the contractor to produce the delay times required in Table I (See 6.2 and 6.3).

TABLE I

Temperature	Delay Time (Seconds) (See 6.4)
-65° to -70°F	0.90 max.
65° to 75°F	0.75 ± .10
160° to 165°F	0.60 min.

3.5 Workmanship.-Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.5.1 Metal defect.-The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.5.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item, or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with MIL-A-2550.

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4.2 First article sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 23 delay bodies and twenty-three primer retainers for protective finish tests and 30 delay element assemblies for ballistic tests as indicated in Table II and 10 sets of unassembled components (including explosive components) for dimensional and visual examination.

TABLE II

Paragraphs Requirement	Test	Test	Sample Size	Type of Test Destr.	Non-Destr.
3.2	4.4.1	Protective Finish	23 delay bodies 23 primer retainers	X	-
3.3	4.4.2	Vibration	12 Cartridges from Ballistic Sample	-	X
3.4.2	4.4.3.1	Ballistics	30 delay elements	X	-

The sample shall be manufactured using the same materials, equipment, process and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.-After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.-The sample for ballistic tests shall be temperature conditioned and tested to determine compliance with Table I. The sample for firing shall consist of the following:

TABLE III

Temperature	Vibrated	Not Vibrated
-65° to -70°F	4	6
65° to 75°F	4	6
160° to 165°F	4	6

4.2.1.3 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

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4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.-A lot shall consist of an identifiable group of metal parts assemblies consisting of

a. Components produced by a homogeneous process
b. Delay charges from not more than one lot
c. Primers from not more than one lot

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristic) the criteria in Table IV shall be used. For total defects within a major or minor class, the criteria in Table V shall be used. All non-conforming material shall be rejected. Four power (4X) magnification may be used to perform visual examination for such defects as burr and foreign matter. The roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination.

TABLE IV

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE V

Sample Size for Major Defects	Total Allowable Defects Per Sample Size			
Number of Defects in Class	110	150	225	300
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14

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4.3.2 (Cont'd)

TABLE V (Cont'd)

Sample Sizes for Minor Defects	Total Allowable Defects per Sample Size			
Number of Defects in Class	110	150	225	300
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Body, Delay, Drawing C8594081, covering a detail of Drawing D10522708.

	Defect	Method of Inspection
CRITICAL		
1.	Diameter of flash hole	Gage
2.	Metal defective	Visual
MAJOR		
101.	Pitch diameter of thread, min.	Gage
102.	Major diameter of thread, min.	Gage
103.	Concentricity of body with pitch diameter of thread	Gage
104.	Diameter of thread relief	Gage
105.	Diameter of step at thread relief	Gage
106.	Diameter of cavity, max.	Gage
107.	Diameter of body, min.	Gage
108.	Depth of cavity, max.	Gage
109.	Length from cavity face to body diameter	Gage
110.	Thread damaged or not full	Visual
111.	Protective finish inadequate or defective	Visual
112.	Burr	Visual
113.	Foreign matter	Visual

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4.3.2.1 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Pitch diameter of thread, max.	Gage
202. Diameter of cavity, min.	Gage
203. Diameter of body, max.	Gage
204. Diameter of wrench holes	Gage
205. Diameter of thread pilot	Gage
206. Depth of cavity, min.	Gage
207. Depth of wrench holes	Gage
208. Total length	Gage
209. Length of thread pilot	Gage
210. Length of thread relief	Gage
211. Length of step at thread relief	Gage
212. Location of wrench holes	Gage
213. Radius or chamfer missing or incorrect	Visual
214. Surface finish improper	Visual

4.3.2.2 Gasket, Aluminum, Drawing B8594082, covering a detail of Drawing D10522708.

CRITICAL

None defined

MAJOR	
101. Diameter, outside, min.	Gage
102. Diameter, inside, max.	Gage
103. Metal defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter, outside, max.	Gage
202. Diameter, inside, min.	Gage
203. Thickness	Gage
204. Surface finish improper	Visual

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4.3.2.3 Retaining Cup, Drawing B8597232, covering a detail of Drawing D10522708.

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of counterbore with small hole	Gage
102. Concentricity of body with small hole	Gage
103. Diameter of counterbore, max.	Gage
104. Diameter of small hole, max.	Gage
105. Diameter of body, min.	Gage
106. Depth of counterbore	Gage
107. Surface finish of small hole and body improper	Visual
108. Burr or sharp corner	Visual
109. Foreign matter	Visual
MINOR	
201. Diameter of counterbore, min.	Gage
202. Diameter of small hole, min.	Gage
203. Diameter of body, max.	Gage
204. Total length	Gage
205. Surface finish improper except as otherwise classified	Visual

4.3.2.4 Plug, Firing, Drawing B8597234, covering a detail of Drawing D10522708.

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of flange with body	Gage
102. Diameter of flange, min.	Gage
103. Diameter of body, min.	Gage
104. Radius of striker incorrect	Gage
105. Length of striker	Gage
106. Surface finish on body, improper	Visual
107. Burr or sharp corner	Visual
108. Foreign matter	Visual

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter of flange, max.	Gage
202. Diameter of body, max.	Gage
203. Total length	Gage
204. Length of body	Gage
205. Radius missing or incorrect, except as otherwise classified	Visual
206. Surface finish improper, except as otherwise classified	Visual

4.3.2.5 Primer, Percussion, M42, Drawing B8799925, covering a detail of Drawing D10522708.

CRITICAL	
1. Improper assembly	Visual
2. Any part missing	Visual
3. Primer pellet broken, cracked or dislodged	Visual
4. Metal defective	Visual
5. Primer foil discolored	Visual
6. Mixed primers	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.3.2.6 Primer, Retainer, Drawing C10520003, covering a detail of Drawing D10522708.

CRITICAL	
1. Depth of large counterbore	Gage
2. Depth of primer pocket	Gage
3. Depth of small counterbore	Gage
4. Metal defective	Visual

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4.3.2.6 (Cont'd)

Defect		Method of Inspection
MAJOR		
101.	Pitch diameter of thread, max.	Gage
102.	Minor diameter of thread, max.	Gage
103.	Concentricity of flange with primer pocket	Gage
104.	Concentricity of flash hole with primer pocket	Gage
105.	Concentricity of large diameter with primer pocket	Gage
106.	Concentricity of large counterbore with primer pocket	Gage
107.	Concentricity of small counterbore with primer pocket	Gage
108.	Concentricity of body with primer pocket	Gage
109.	Concentricity of pitch diameter of thread with body	Gage
110.	Symmetry of wrench flats with pitch diameter of thread	Gage
111.	Diameter of body, min.	Gage
112.	Diameter of primer pocket, min.	Gage
113.	Diameter of small counterbore, min.	Gage
114.	Diameter of large counterbore	Gage
115.	Depth of primer pocket	Gage
116.	Depth of small counterbore	Gage
117.	Length of effective thread, min.	Gage
118.	Location of flange	Gage
119.	Location of datum diameter	Gage
120.	Thread damaged or not full	Visual
121.	Protective finish inadequate or defective	Visual
122.	Surface finish or primer pocket improper	Visual
123.	Burr or sharp corner	Visual
124.	Foreign matter	Visual
MINOR		
201.	Pitch diameter of thread, min.	Gage
202.	Diameter of body, max.	Gage
203.	Diameter of primer pocket, max.	Gage
204.	Diameter of small counterbore	Gage
205.	Diameter of flange	Gage
206.	Depth of wrench flats	Gage
207.	Depth to bottom of threaded cavity	Gage
208.	Length of wrench flats	

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
MINOR (Cont'd)	
209. Length across wrench flats	Gage
210. Total length	Gage
211. Location of wrench flats	Gage
212. Radius or chamfer missing or incorrect	Visual
213. Surface finish improper, except as otherwise classified	Visual

4.3.2.7 Assembly, Drawing D10522708.

CRITICAL	
1. Improper assembly	X-ray 1/ and Visual
2. Any part missing	X-ray 1/ and Visual
3. Retainer cup not flush or below with surface "S"	Visual
4. Firing plug not flush or below with retaining cup	Visual
5. Primer not seated to full depth of primer pocket	Visual
6. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Concentricity of flange with body	Gage
102. Junction of delay body, gasket and primer retainer not properly sealed	Visual
103. Primer not properly crimped and sealed	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter of body, max.	Gage
202. Total length, max.	Gage

1/ The delay element shall be X-rayed with the element in a position horizontal to the rays of the X-ray machine. The X-rays properly identified shall be forwarded to the procuring activity.

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4.3.2.8 Ammunition Container, Metal, M21, Drawing C8864162.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual/Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat seal incomplete, loose or distorted	Visual
205. Cut or damaged through impregnated box	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.9 Box, Packing, Ammunition for Delay Element, M5 or M8, Drawing C8826729.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual/Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

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4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.-Twenty-three delay bodies and 23 primer retainers shall be selected for protective finish tests. Failure of any component to comply with the requirements of 3.2 shall be cause for rejection of the lot.

4.3.3.2 Vibration.-Twelve of the 30 assembled X-rayed delay elements of the initial production sample shall be subjected to vibration in accordance with 3.4. After vibration, the 12 vibrated delay elements shall be X-rayed as directed in 1/ of 4.3.2.7. Failure of any delay element to comply with the requirements of 3.3 shall be cause for rejection of the lot. When there is a change in the method of charging the delay element, 12 additional delay elements shall be selected and tested as indicated herein.

4.3.3.3 Functional.-The following test shall be performed by the Government:

4.3.3.3.1 Ballistics.-Thirty test delay elements shall be grouped and temperature conditioned in accordance with 4.4.3.1.1 and Table VI. Failure of any delay element to comply with the requirements of 3.4.2 shall be cause for rejection of the lot.

TABLE VI

Temperature	Sample Size
-65° to -70°F	10
65° to 75°F	10
160° to 165°F	10

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL-10522708.

4.4 Test methods and procedures.

4.4.1 Protective finish.-Test procedures for protective finish tests shall be in accordance with MIL-C-60539. The cartridge and primer retainer shall be sectioned longitudinally to better expose the critical surfaces to the salt spray.

4.4.2 Vibration.-The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

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4.4.3 Functional.

4.4.3.1 Ballistics.-Delay elements of the sample shall be grouped and temperature conditioned in accordance with 4.4.3.1.1 and rigidly supported in the test fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be reconditioned in accordance with 4.4.3.1.1.

4.4.3.1.1 Temperature conditioning.-A standard inert delay element containing a temperature sensitive device shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of one (1) hour longer than the time necessary to bring the device contained in the inert delay element to the required temperature. The vent holes in the delay elements shall be sealed against the environmental atmosphere of the conditioning chamber during the conditioning period. Upon completion of the temperature conditioning, the vent seal shall be removed.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be in accordance with Drawings C8826729 and C8864162, except that the markings shall refer to delay element XM44, Assembly.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Advisory output material (formation).-Approximately 23.7 percent Boron, 70.7 percent potassium nitrate and 5.6 percent binder. Advisory individual delay element output material weight - approximately 1.5 grains.

6.3 Advisory delay charge composition (formulation) approximately 95 percent barium chromate and 5 percent boron. Advisory individual delay element charge weight: approximately 18 grains.

6.4 Delay time is defined as the time lapse between the impact of the firing pin on the primer and evidence of gas flow from the vent hole of the delay element.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A295

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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